Sanaul Haque

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Assignment 108

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IARCO RESEARCH PROPOSAL

Full Legal Name: Riana Agarwal

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Research Topic: Development of a Low-Cost Smart Saturation Detection System for Sanitary Pads and Adult Diapers Using Embedded Conductive Sensor Layers and Wireless Caregiver Alerts

Title: Smart PeriodCare - Conductivity-Based Saturation Detection and AI-Powered Pad Change Prediction

Research Problem

The timely replacement of sanitary pads or incontinence pads is essential to maintain hygiene, comfort, and protect the skin. Yet, manual detection of wetness can prove to be difficult and tedious, particularly among young girls, older adults, patients in bed, babies, and those with disabilities. Young patients might face difficulties in comprehending the rate at which the menstrual flow accumulates, which results in anxiety, embarrassment, clothes getting stained, and lack of self-confidence in public places such as schools and colleges. Currently available smart hygienic systems rely on either humidity, textile or other kinds of sensors; however, their complexity may lead to issues like false positive results, sensor location, and user discomfort. This project seeks to create a simple and inexpensive conductance-based sensor system for detecting the presence and saturation of fluid in the particular section of the sanitary product. Besides the real-time detection of product wetness, the prediction system based on AI could help determine the appropriate moment for changing the product.

Existing Literature

Studies on smart hygienic products have been interested in the application of electronic sensors to detect moisture and incontinence. Jeong and Park [1] reviewed the various types of sensors used in urinary containment products and recommended that the application of sensors in monitoring would help minimize labor in incontinence care. Ben Arous et al. [2] reviewed non-invasive wearable sensors for detecting urinary incontinence and mentioned the use of conductive sensors as an important sensing technology. They noted that electrodes were able to detect the presence of fluid by detecting changes in resistance, voltage or current levels, but various issues like sensor location, configuration of electrodes, false alarms, portability and comfort need to be taken into account. Tekçin et al. [3] showed a flexible humidity sensor built into a disposable diaper. Tekcin Meltem and Senem Kursun [4] tested textile moisture sensors for smart diapers with water and artificial urine solutions. Wang Ziheng et al. [5] demonstrated in-pad biosensors which were able to detect biomarker content and moisture levels in artificial vaginal fluid. Based on the earlier research, the current project will make use of conductivity to determine the presence of fluid, specifically evaluating the configuration, position, and composition of electrodes created in the course of prototype creation. The need to make the apparatus portable and comfortable, as well as to ensure that its use does not create additional burden on the caregivers, has been addressed by creating a prototype combining reusable and disposable elements using a small wireless controller ESP32-C6 and wireless alert system. Also, the project expands the existing approaches involving the use of sensors by combining real-time detection of bedwetting with an automatic indication of the need to change the pad.

Research Question

Is it possible to develop an inexpensive embedded conductivity sensor that can accurately identify saturation points in feminine sanitary pads and use its readings in an AI predictive model to give out personalized pad change reminders in real-time?

Methodology

In the Smart PeriodCare system, there will be a hybrid of disposable and reusable technology that integrates conductivity saturation sensors, wireless alerts, and AI for predicting when it is time for the pad change. There will be three steps in the methodology: conductivity measurement, generation of the wireless alert, and prediction. In the physical system, there will be conductive electrodes, XIAO ESP32-C6 transmitter and a second ESP32-C6 receiver to get visual and tactile alerts respectively. In the AI process, the user inputs and synthetic data will serve as the education proof of concept.

Conductivity Measurement and Saturation Detection: Change in conductivity due to presence of the conductive liquid around the electrodes in the absorbent pad will be detected by the system. Resistance is high in absence of the liquid while conductivity is increased upon reaching the fluid the electrodes. The system is designed to detect whether the fluid has reached the sensor area instead of measuring the quantity of absorption.

Electrode Development: Copper and nichrome electrodes will be tested as conductive materials to evaluate their suitability for saturation detection. Different electrode shapes, dimensions, lengths, and positions will be tested on sanitary pads to determine the configuration that provides reliable and accessible sensing. The planned research execution will follow a systematic sequence:

Different electrode positions will be tested on XL, XXL and L size sanitary pads to identify suitable sensing regions

Each sanitary napkin will be tested using 200 mL of test fluid to study the sensing behaviour over a larger absorbent area.

A sanitary napkin will be mounted on a curved structure and tested with blood replication of test fluid to investigate electrode performance.

The electrode dimensions will be reduced and tested on sanitary pads to determine how electrode size influences detection behaviour and sensing coverage.

Different electrode lengths will be compared to identify an appropriate electrode length for reliable detection.

Different electrode positions will be evaluated to determine which placement provides consistent contact with the test fluid while remaining practically accessible for integration into the pad.

The results from the above experiments will be compared to identify the most suitable combination of electrode material, length, dimensions, and placement.

Blood Replication Fluid: The test fluid to be used in the experiments is not menstrual blood but a blood simulant fluid. This fluid is made up of glycerol and water mixed in a ratio of 1:5 respectively, to give appropriate conductivity and flow rate. The total amount is 120 ml.

Electrical Measurement: The output of the electrode will be connected to the analog input of ESP32 and the ADC value from 0 to 4095 will be acquired with the help of 12-bit ADC. An empirical value of around 2500 for ADC will be chosen as the threshold value to determine whether the material is dry or conductive.

Sensor condition	ADC response	Interpretation

Dry	<2500	Low conductivity
Conductive/wet	>2500	Increased conductivity

Detection in Real-Time: ESP32 will be measuring conductivity values and comparing them against certain threshold levels, thus providing automated monitoring of the pads without human control.

Two-Node Architecture: There will be two units of XIAO ESP32-C6 to isolate the disposable sensing pad from the notification device. The first one – the sender node – will attach to the conductive electrodes on the sanitary pad for:

1. Measuring conductivity values.
2. Determining the current level of saturation.
3. Interpreting the sensor reactions in terms of saturation.
4. Sending this information to the receiver node wirelessly.

The second node is designed as the notification device and its job is to receive information about the current level of saturation and provide notifications of different kind.

Wireless Communication: Two ESP32-C6 boards will transmit wirelessly through ESP-NOW protocol. As a result, there will be fast data transmission of information about the pad's saturation, sensors, and voltage. Having a different receiver will be convenient in creating a reusable and wireless alert system.
The receiver will have a system of progressive notifications: a yellow LED light will signal the first stage; then there will be a red LED light and a vibration engine at the higher level of pad saturation. In other words, the notifications will be both perceivable and discreet.

AI-Based Pad-Change Prediction:The third module will contain an AI algorithm that predicts whether the pad should be replaced based on users' and the pad's features. This module will be used for programming purposes only without the possibility to make a medical diagnosis.
Inputs of the algorithm include age, pad type, pad size, flow type, time since the previous replacement, and cycle day. Types of pads like SAP, fluff pulp, air laid paper, and cotton will be considered to describe the construction of pads.

Dataset Preparation: For testing and developing the AI module, a synthetic dataset will be created which will have the necessary input variables and the following three target classes: No Change Needed, Change Soon, and Change Recommended. For categorical inputs like pad type, size, and flow type, numerical encoding will be done. Data processing and preparation will be done using Pandas while numerical encoding and model training will be done using Python.

Random Forest Model: As for the prediction part, a Random Forest Classifier will be used to predict whether a change in the pad is needed based on the input variables provided by the user. This classifier will consist of many decision trees that will help in classifying the need for a change in the pad from six user and usage variables. It will

produce any one of the three recommendations: No Change Needed, Change Soon, or Change Recommended. The trained model will be saved using Joblib.

Prediction Output: The outputs of the algorithm can fall into one of the following categories:

1. No change required: The current input scenario implies that there is no immediate need for change.
2. Change soon: The algorithm recommends monitoring the pad and getting ready for the change.
3. Change recommended: According to the input provided, the algorithm recommends changing the pad.

In addition to this, users will also be able to see the confidence level of the algorithm's prediction along with the predicted time-based output. It is crucial to understand that the outputs generated by the system are demo outputs and cannot be taken as clinically validated recommendations.

Research Topic

In this research paper, emphasis is put on the design of low cost smart hygiene monitoring technology through the use of conductivity sensors, electronics, and wireless technology. The greatest achievement of this technology is in the splitting of the whole technology into two parts, which include the disposable and electronics controllers. Conductive electrodes may be embedded in the absorbent material, while the ESP32-C6 electronics may be taken out and reused. This is because of the limitations that are posed by disposability of the smart hygiene device.

The research is particularly relevant to:

1. Menstrual hygiene
2. Pediatric care
3. Elderly care
4. Incontinence management
5. Bedridden patient care
6. Assisted living
7. Cognitive-disability care
8. Smart healthcare
9. Wearable/embedded sensing
10. IoT-enabled healthcare

The design also includes an element that is important from a human point of view. For instance, the design includes the fact that a young lady experiencing her first menstruation has to cope with shame, fear, avoiding some activities, and also fearing involvement in school activities as discussed by Vashisht et al. [6]

Quality of Writing

The use of scholarly and professional language is evident in the formulation of this proposal. The amount of jargon is kept at a bare minimum, and the structuring of the proposal adheres to the standards set for research writing. The proposal adheres to all grammatical and formatting conventions.

References

[1] Jeong, Gyeonghui, and Sangsoo Park. “Review of Urinary Continence Care Products Using Sensor Technology to Improve Effectiveness.” Proceedings of the Institution of Mechanical Engineers, Part H: Journal of Engineering in Medicine, vol. 233, no. 1, 2019, pp. 91-99.

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[3] Tekçin, Gökçe, et al. “Wearable and Flexible Humidity Sensor Integrated to Disposable Diapers for Wetness Monitoring and Urinary Incontinence.” Electronics, vol. 11, no. 7, 2022, article 1025.

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[6] Vashisht, Aditi, et al. "School absenteeism during menstruation amongst adolescent girls in Delhi, India." Journal of family & community medicine 25.3 (2018): 163.